Work in progress — draft, not peer reviewed

# A Lean-Verified Conditional Reduction of Periodic Navier–Stokes Regularity on $\mathbb{T}^3$ to a Single Integrability Hypothesis

*Conditional results; draft — public repository, not peer reviewed*

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**Scope.** This paper reports a *conditional* result. It does **not** solve the Clay Millennium Problem, and it does not prove global regularity of the 3D Navier–Stokes equations. The single remaining hypothesis, `PeriodicBKMIntegrability`, is a named Prop that is **not proved**. It is equivalent in difficulty to Clay Statement (B).

**How to cite / What is claimed.** In Lean 4 with Mathlib, we give a machine-checked reduction of Clay Millennium Statement (B) (periodic Navier–Stokes on $\mathbb{T}^3$) to the single hypothesis `PeriodicBKMIntegrability`. We prove this hypothesis is equivalent to global existence of smooth solutions for every smooth periodic initial field. The BKM criterion and $H^3$ control are proved unconditionally, using only the standard axioms. `PeriodicBKMIntegrability` remains open. The equivalence of vorticity integrability with continuation of smooth solutions is classical (Beale–Kato–Majda, 1984 [1]); the novelty here is the complete, `sorry`-free formalization. Cite as: B. S. Frohman, reduction paper, DOI 10.5281/zenodo.23842296; main paper, DOI 10.5281/zenodo.23842556.

**Abstract**

In the Lean 4/Mathlib [4, 12] repository `BenFrohman/periodic-ns-classical-continuation` (commit `f9a26c7`), we prove two results with no hypothesis Props:

- $H^3$ Beale–Kato–Majda control on $\mathbb{T}^3$, `periodicBKMHsControl_three`;
- the periodic BKM blow-up criterion [1], `periodicBKMCriterion`.

We deduce the exact periodic Clay Statement (B) [5], `Clay_B_exact`, *conditionally* on the single named Prop `PeriodicBKMIntegrability` (`clay_B_exact_of_integrability`). The axioms used are only `propext`, `Classical.choice` and `Quot.sound`.

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*Keywords.* Navier–Stokes equations; torus; Beale–Kato–Majda criterion; conditional regularity; Lean 4; formal verification; work in progress.

## 1 Introduction

The Beale–Kato–Majda criterion [1] asserts that a smooth solution of the three-dimensional Navier–Stokes equations can cease to be smooth at a finite time $T$ only if $\int_0^T \|\omega(t)\|_{L^\infty}\,\mathrm{d}t = \infty$. Consequently, Statement (B) of the Clay Millennium Problem [5] on the torus is equivalent to the time-integrability of the vorticity sup-norm for every smooth periodic initial field. In this paper we report a machine-checked version of this reduction in Lean 4: the criterion


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and the required $H^3$ control are proved with no hypothesis Props, and Statement (B) is derived from the single named hypothesis `PeriodicBKMIntegrability`, which remains open. Section 3 tabulates precisely what is stated and what is proved.

## 2 The periodic chain on $\mathbb{T}^3$

Throughout, $\nu > 0$, $u_0$ is smooth, periodic and divergence-free, and $(u, p)$ is the maximal smooth solution of

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \qquad \operatorname{div} u = 0, \qquad u(0) = u_0 \quad \text{on } \mathbb{T}^3, \tag{2.1}$$

with existence time $T^*$; $\omega = \operatorname{curl} u$. The definitions `Clay_B_exact`, `PeriodicBKMCriterion` and `PeriodicBKMIntegrability` come from the dependency `fefferman-clay-navier-stokes`.

**Named hypothesis 2.1** (`PeriodicBKMIntegrability` $\nu$; **open**). For every maximal solution and every $T$, $t \mapsto \|\omega(t)\|_{L^\infty}$ is integrable on $[0, T]$:

$$\int_0^T \|\omega(t)\|_{L^\infty} \, \mathrm{d}t < \infty. \tag{2.2}$$

**Theorem 2.2** (`periodicBKMHsControl_three`†; unconditional). *For every* $\nu$, `PeriodicBKMHsControl` $\nu$ 3 *holds. That is,*

$$T^* = T < \infty \quad \textit{and} \quad \int_0^T \|\omega\|_{L^\infty} \, \mathrm{d}t < \infty \quad \Longrightarrow \quad \sup_{t \in [0,T)} \|u(t)\|_{H^3} < \infty.$$

*Proof route (all formalized).* The $H^3$ energy identity reduces the claim to a commutator bound and a low-frequency vorticity bound.

- The $s = 3$ commutator bound [6] is

  $$-2 \sum_i \int \left(1 - \frac{\Delta}{4\pi^2}\right)^3 v_i \left((v \cdot \nabla)v\right)_i \, \mathrm{d}x \le C \|\nabla v\|_{L^\infty} \|v\|_{H^3}^2$$

  (`periodicPhysicalCommutatorBound`†, `CommutatorAssembly.lean`).

- The low-frequency bound is

  $$\|V_R \nabla v\|_{L^\infty} \le K(1 + \log R) \|\omega\|_{L^\infty} \qquad (R \ge 2)$$

  (`periodicLowFreqVorticityBound`†, `LowFreqAssembly.lean`). It is proved through the Riesz form of Biot–Savart, a smooth dyadic decomposition with uniformly $L^1$-bounded block kernels, and a uniform $L^1$ bound for the de la Vallée Poussin kernel.

Grönwall's inequality with the log splitting then closes the bound.

**Theorem 2.3** (`periodicBKMCriterion`†; unconditional). *For every* $\nu$, `PeriodicBKMCriterion` $\nu$ *holds. If* $\nu > 0$ *and* $T^* = T < \infty$, *then*

$$\int_0^T \|\omega(t)\|_{L^\infty} \, \mathrm{d}t = \infty.$$

**Theorem 2.4** (`clay_B_exact_of_integrability`†; conditional). `PeriodicBKMIntegrability` $\nu \Rightarrow$ `Clay_B_exact` $\nu$.


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*Remark* 2.5 (Equivalence in difficulty). The converse direction also holds. If Statement (B) gives a global smooth solution, then by uniqueness the maximal solution is global and smooth, so $\|\omega\|_{L^\infty}$ is continuous and integrable on every $[0, T]$. Hence `PeriodicBKMIntegrability` is *equivalent* to the periodic Clay Statement (B); it is not a weaker, merely technical residue. This converse, from Statement (B) itself, is argued informally here. What *is* proved in Lean (`periodicBKMIntegrability_iff_globalLifespan`, `IntegrabilityAttack.lean`) is that `PeriodicBKMIntegrability` is equivalent to `PeriodicGlobalLifespan`, i.e. $T^* = \infty$ for all smooth data. Its proof uses `integrableOn_vorticity_of_lt_tstar`: $\int_0^T \|\omega\|_{L^\infty}\,\mathrm{d}t < \infty$ whenever $T < T^*$. This paper is therefore a verified reduction, not a proof of regularity.

## 3 Stated vs. proved

| Statement | Status |
|---|---|
| $H^3$ BKM control, `periodicBKMHsControl_three` | proved unconditionally |
| Commutator bound ($s = 3$), `periodicPhysicalCommutatorBound` | proved unconditionally |
| Low-frequency vorticity bound, `periodicLowFreqVorticityBound` | proved unconditionally |
| Periodic BKM criterion, `periodicBKMCriterion` | proved unconditionally |
| Clay Statement (B), via `clay_B_exact_of_integrability` | proved conditionally on `PeriodicBKMIntegrability` |
| `PeriodicBKMIntegrability` | not proved (open; equivalent to Clay (B)) |
| Converse: (B) ⇒ `PeriodicBKMIntegrability` | not proved in Lean (informal argument only) |
| `periodicBKMIntegrability_iff_globalLifespan`† | proved unconditionally |
| `integrableOn_vorticity_of_lt_tstar`† (below $T^*$) | proved unconditionally |
| Small data, `integrability_of_smallDataGlobal`† | proved from named `PeriodicSmallDataGlobal` (not formalized) |
| Prodi–Serrin, `integrability_of_prodiSerrin`† | proved from named `PeriodicProdiSerrinCriterion` (not formalized) |
| $\kappa = 0$ route, `periodicBKMIntegrability_of_vorticityAPriori`† | proved from `PeriodicVorticityAPrioriBound` (open) |
| $H^1$ enstrophy identity, `periodicEnstrophyIdentity`† | proved unconditionally |
| Small zero-mean $H^1$ data, `periodicTstar_eq_top_of_smallH1_of_rate`† | proved from named `PeriodicEnstrophyRateBound` + `PeriodicH1BlowupCriterion` (not formalized; [7, 8]) |
| Ladyzhenskaya-type trilinear bound, `periodicH1TrilinearBound`† | proved (explicit unoptimized constant; Fourier/Agmon route; the inequality is classical: Ladyzhenskaya 1958 (2D), 1961 monograph (3D) [7]) |
| Mean conservation, `periodicMeanConservation`† | proved unconditionally |
| Small zero-mean $H^1$ data, `periodicTstar_eq_top_of_smallH1_ladyzhenskaya`† | proved from `PeriodicH1BlowupCriterion` only (not formalized; [8]) |


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| | |
|---|---|
| Earlier small-$H^1$ route, `integrability_of_smallH1`† | superseded: its hypothesis `PeriodicEnstrophyRiccati` ($\alpha > 0$) fails for constant flows (bug fixed in `ea5133d`) |
| Tether cubic rate, `clay_B_exact_of_tetherCubicRate`† | proved from `PeriodicTetherCubicRate` (open) |
| Uniform cubic rate vs. scaling, `uniform_cubic_rate_false_of_scaling`† | proved (abstract scaling step) |
| Direction route, `periodicBKMIntegrability_of_direction`† | proved from `PeriodicDirectionCriterion` (not formalized) + `PeriodicDirectionRegularity` (open) |
| Direction criterion, `periodicDirectionCriterion_of_growth`† | proved from `PeriodicDirectionEnstrophyGrowth` (CF 1993 [3] / BdV–B 2002 [2], open here) + `PeriodicH1BlowupCriterion` (Leray, open here) |
| Direction criterion ($L^1$ rate), `periodicDirectionCriterion_of_growthL1`† | proved from `PeriodicDirectionEnstrophyGrowthL1` (open here) + `PeriodicH1BlowupCriterion`; CF depletion factor `abs_cfDepletion_le` and Grönwall `gronwall_integrating_factor` proved |
| Bending-rate identity, `hasDerivAt_vortDir_lagrangian`† | proved: $D\mathbf{t}/Dt = (I - \mathbf{t}\otimes\mathbf{t})S\mathbf{t} + \nu\lvert\omega\rvert^{-1}(I - \mathbf{t}\otimes\mathbf{t})\Delta\omega$ |
| Hidden property, `PeriodicBendingRateHiddenProperty`† | **open** (new formulation dated Oct 31 2026; precursors (geometric only, no regularity claims): the author's May 17/21 2026 notes ($\Pi_u$ orthogonal to $u$, "invisible to the dynamics generated by $H$") and Aug 10 2026 notes *The Frohmanian Symplectic Tether* ($B_{\text{anti}}$ via $\hat{u}\times(\cdot)$, $B_{\text{sym}}$ with pointwise $\Pi_u$)); proved to imply `PeriodicDirectionRegularity` |
| Leray criterion, `periodicH1BlowupCriterion_of_smoothing`† | proved from `PeriodicH2SmoothingRate` + `PeriodicH3SmoothingRate` (open here) and the proved $H^3$ blow-up alternative |
| Energy equality, `periodicEnstrophyTimeIntegrable_of_cancellation`† | proved: `PeriodicEnstrophyTimeIntegrable` from `PeriodicNonlinearEnergyCancellation` (open here; $\langle(u\cdot\nabla)u, u\rangle = 0$ [8, 10]) |
| Leray criterion, standard form, `periodicH1BlowupCriterion_of_smoothingStd`† | proved from `PeriodicH2SmoothingRateStd` + `PeriodicH2RootTimeIntegrable` + `PeriodicH3SmoothingRate` (open here [11, 9]); the pointwise `PeriodicH2SmoothingRate` is stronger than the standard estimate |
| $L^1$ growth, `periodicDirectionEnstrophyGrowthL1_of_holderStretching`† | proved from `PeriodicHolderStretchingEstimate` + `PeriodicEnstrophyTimeIntegrable` (open here) |
| $\kappa = 0$ tether, `tetherImpliesDirection_zero_iff`† | proved: headline Prop at $\kappa = 0 \Leftrightarrow$ `PeriodicDirectionRegularity` |
| Clay problem / NS global regularity | not proved, not claimed |

# A Theorem-to-Lean map

Names marked † were coined during formalization and are not the author's terminology; they may be renamed later.

| Lean name | File | Repo @ commit |
|---|---|---|
| `periodicBKMHsControl_three`† | `PeriodicNSContinuation/BKMFinal.lean` | pnscc @ `ea5133d` |


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| | | |
|---|---|---|
| periodicBKMCriterion† | PeriodicNSContinuation/BKMFinal.lean | pnscc @ ea5133d |
| clay_B_exact_of_integrability† | PeriodicNSContinuation/BKMFinal.lean | pnscc @ ea5133d |
| periodicBKMHsControl_three_of_lowFreq† | PeriodicNSContinuation/CommutatorAssembly.lean | pnscc @ ea5133d |
| periodicPhysicalCommutatorBound† | PeriodicNSContinuation/CommutatorAssembly.lean | pnscc @ ea5133d |
| periodicLowFreqVorticityBound† | PeriodicNSContinuation/LowFreqAssembly.lean | pnscc @ ea5133d |
| periodicBKMCriterion_of_hsControl | PeriodicNSContinuation/Continuation.lean | pnscc @ ea5133d |
| PeriodicBKMHsControl† (def) | PeriodicNSContinuation/Continuation.lean | pnscc @ ea5133d |
| clay_B_exact_of_bkm† | PeriodicNSContinuation/LocalExistenceProof.lean | pnscc @ ea5133d |
| periodicBKMIntegrability_iff_globalLifespan† | PeriodicNSContinuation/IntegrabilityAttack.lean | pnscc @ ea5133d |
| periodicEnstrophyIdentity† | PeriodicNSContinuation/EnstrophyIdentity.lean | pnscc @ ea5133d |
| periodicTstar_eq_top_of_smallH1_of_rate† | PeriodicNSContinuation/EnstrophyIdentity.lean | pnscc @ ea5133d |
| periodicH1TrilinearBound† | PeriodicNSContinuation/Ladyzhenskaya.lean | pnscc |
| periodicMeanConservation† | PeriodicNSContinuation/MeanConservation.lean | pnscc |
| clay_B_exact_of_tetherCubicRate† | PeriodicNSContinuation/TetherClaimAttack.lean | pnscc @ ea5133d |
| periodicBKMIntegrability_of_direction† | PeriodicNSContinuation/TetherClaimAttack.lean | pnscc @ ea5133d |
| tetherImpliesDirection_zero_iff† | PeriodicNSContinuation/TetherKappaZero.lean | pnscc @ ea5133d |
| Clay_B_exact (def) | FeffermanClayNS/FeffermanStatements.lean | dependency |
| PeriodicBKMCriterion, PeriodicBKMIntegrability (defs) | FeffermanClayNS/HighSetDepletionAndBKM.lean | dependency |

pnscc = `BenFrohman/periodic-ns-classical-continuation` (public). "dependency" = `fefferman-clay-navier-stokes`, as pinned in the lake manifest.

## B Axiom audit

At the current commit, `lake build` succeeds with 0 `sorry`, and `lake env lean AxiomAudit.lean` prints "AxiomAudit PASS: only propext, Classical.choice, Quot.sound". Every theorem above is in the audit, including those of `SmallDataAttack`, `TetherClaimAttack`, `EnstrophyIdentity` and `TetherKappaZero` (commits `052cef1`, `ea5133d`). The audit also covers `DirectionCriterion`, `BendingRate` and `ClassicalSteps`: the $L^1$ direction criterion `periodicDirectionCriterion_of_growthL1`, the bending-rate identity `hasDerivAt_vortDir_lagrangian`, the depletion bound `abs_cfDepletion_le`, the reductions `periodicDirectionRegularity_of_bending` (from the open bending-rate hypothesis `PeriodicBendingRateHiddenProperty`), `periodicH1BlowupCriterion_of_smoothing` and `periodicDirectionEnstrophyGrowthL1_of_holderStretching`; all use only `propext`, `Classical.choice`, `Quot.sound` (current commit; 0 `sorry`).


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**Counting convention.** Two counts are reported, each with its method. (i) *Source count:* 726 declarations introduced by the keywords `theorem` or `lemma` (optionally preceded by attributes or modifiers) in the tracked `.lean` files of this repository, excluding `AxiomAudit.lean`. (ii) *Kernel count:* 833 non-internal theorem constants in the `PeriodicNSContinuation` module tree, as reported by `AxiomAudit.lean`; this includes theorems generated automatically by Lean. An earlier figure of 727 was not obtained by either method and is withdrawn; the figure 726 used in versions $\leq 7$ is the source count.

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